PRD

Embedded Weather Station and User Interface System.

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The why of the product.

This product is primarily used to keep track of the weather and the atmospheric pressure and display this data to a screen.

Secondary features may include but not be limited to flashing displays and lights from libraries .

Product users.

Product users include any person who wishes to track weather data and atmospheric pressure such as airlines.

Goals and success metrics.

At a minimum the goal for this product is the display the current weather and atmospheric pressure on a display with an interactive menu navigated via a joystick. Allowing users to switch between different data visualization pages including pressure in hectopascals, pressure in millibars in a large font format, and temperature in Celsius.

Other possible features include but not limited to (at this time):

Altitude reading.

Advanced graphical interface.

Cloud data storage.

Scope (in scope/out of scope)

Functional and non-functional requirements.

Functional:

- The device must read in temperature data
- Read barometric pressure data.
- Stream data from met erinn on sea level.

Non-Functional:

- Resists extreme temperature change - 20 degrees or more in the + or – .
- Uses Wi-Fi and satellite to find sea level .
- Not draw too much power – greater than 13V.
- Be scalable.
- fit into a wide range of products easily and be maintainable.

Assumpions, Risks and constraints

Assumptions:

- Will be using a 12v power - power pack.

Risks

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Constraints

- Power delivery.
- Size.
- Time – one semester to complete.
- Atmospheric conditions affecting transmission.
- Optimal circuit design